

Software Test Plan (STP) – Vehicle Parking System

Project: Vehicle Parking System (Project No. 10) – a system for managing parking slots and vehicle tracking

Version: 1.1

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Related documents: Vehicle Parking System SRS v1.1 (requirement IDs FR-01 to FR-23, NFR-01 to NFR-07, SEC-01 to SEC-07, BR-01 to BR-05).

1. Introduction

Purpose: This document defines the test plan for the Vehicle Parking System (VPS) v1.1. It sets out the objectives, scope, strategy, resources, schedule and responsibilities for testing, and links every requirement of the SRS to at least one test case.

Scope: Testing covers authentication and access control, vehicle registration, parking slot management, vehicle entry and slot allocation, vehicle exit and fee calculation, fee rate configuration, vehicle search/tracking/history, the occupancy report, and the non-functional and security requirements of the SRS. The VPS is a MERN-stack web application (React.js, Node.js/Express.js, MongoDB). Third-party platform internals and hardware are excluded (Section 4).

References: VPS SRS v1.1 (including the Requirements Traceability Matrix in Appendix C); VPS Software Architecture & Design Specification (in preparation); PES University SE Mini-Project Deliverables Part-1; IEEE Std 829-2008 (software test documentation); OWASP Application Security Verification Standard and OWASP Top 10.

Definitions: VPS (Vehicle Parking System), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), TC (Test Case), FR/NFR/SEC/BR (Functional, Non-Functional, Security requirement, Business Rule), UAT (User Acceptance Test), p95 (95th percentile response time), RBAC (Role-Based Access Control), TLS (Transport Layer Security).

2. Test Items

- Authentication and User Management module (login, logout, user and role administration)
- Security middleware (token check, role-based authorization, input validation)
- Vehicle module (registration, search, tracking, history)
- Parking Slot module (slot management, availability display)
- Parking Transaction module (entry, slot allocation, exit, slot release)
- Fee module (hourly rate configuration and fee calculation)
- Report module (occupancy report)
- React.js web user interface
- REST API and MongoDB data layer

3. Features to be Tested

Features are mapped to SRS requirement IDs. Test cases for each requirement are listed in Section 13 and Appendix B.

SRS section / feature	Requirement IDs	Summary of what is verified
5.1 Authentication and Access Control	FR-01 to FR-05	Login, generic error, logout, user and role management, 403 on forbidden functions
5.2 Vehicle Registration	FR-06, FR-07	Registration, number format, rejection of invalid or duplicate input
5.3 Parking Slot Management	FR-08 to FR-10	Add slots, set Unavailable/Available, status and count display
5.4 Vehicle Entry and Slot Allocation	FR-11 to FR-14	Entry recording, allocation by type, no-slot refusal, atomic record with ticket
5.5 Vehicle Exit and Fee Calculation	FR-15 to FR-18	Exit recording, fee with grace period and rounding, summary, slot release
5.6 Fee Rate Configuration	FR-19	Hourly rates; a change applies only to later exits
5.7 Vehicle Search, Tracking and History	FR-20 to FR-22	Search, current status, parking history
5.8 Reports	FR-23	Parking Occupancy Report
6.1 Performance	NFR-01 to NFR-03	Response times at 50 concurrent users (search) and for entry and exit
6.2 Safety and Reliability	NFR-04, NFR-05	Concurrent allocation, persistence of acknowledged operations
6.4 Quality attributes	NFR-06, NFR-07	Usability of entry; browser and screen-size compatibility
6.3 Security	SEC-01 to SEC-07	Authentication, RBAC, password storage, TLS, input validation, session timeout, secrets
6.5 Business Rules	BR-01 to BR-05	Enforced through the functional and concurrency tests above

4. Features Not to be Tested

- Online payment, number-plate recognition, hardware gate control, reservations and native mobile apps (out of scope in SRS Section 1.3) and Vehicle Owner self-service functions (removed from SRS v1.1)
- Internals of third-party software: Node.js, Express.js, React.js, MongoDB and npm libraries (assumed tested by their maintainers; only our use of them is tested)
- Hosting provider infrastructure and network reliability outside the team's control
- Load beyond 50 concurrent users and browsers other than those named in NFR-07

5. Test Approach / Strategy

Levels:

- Unit tests (Jest): validation functions, fee calculation, slot allocation logic, authorization checks.

- Integration tests (Jest + Supertest with a test MongoDB): API ↔ database, module-to-module flows such as entry → slot status.
- System tests (Cypress and Postman/Newman): end-to-end workflows through the UI and API.
- Acceptance tests (UAT): reviewer/instructor walk-through of the main scenarios against the SRS.

Types:

- Functional testing of every FR and business rule, using equivalence partitioning and boundary value analysis for inputs and fee durations.
- Negative testing of invalid, missing and duplicate input and forbidden actions.
- Regression testing: the automated suite is re-run after every defect fix and before each milestone.
- Performance testing with JMeter (response time and load, NFR-01 to NFR-03).
- Concurrency and reliability testing (NFR-04, NFR-05).
- Usability testing with three first-time users (NFR-06).
- Compatibility testing across browsers and screen sizes (NFR-07).
- Security testing (Section 5.1).

Entry criteria: A stable build is deployed to the test environment, smoke test passes, test data is loaded, and test cases are reviewed.

Exit criteria: 100% of planned test cases executed; all High-priority test cases passed; 0 open Critical or High defects; Medium defects have an agreed plan; every requirement in the RTM has at least one executed passing test case; performance targets of NFR-01 to NFR-03 are met.

Pass/fail criteria: A test case passes only when all expected results are observed. Any deviation is logged as a defect with the severity below.

Severity	Definition	Example
Critical	System unusable or data loss/security breach	Two vehicles assigned to one slot; password stored in plain text
High	Major function fails, no workaround	Exit cannot be completed; wrong fee
Medium	Function fails but a workaround exists	Report shows wrong sort order
Low	Cosmetic or minor usability issue	Label typo

5.1 Security Validation

Security tests verify the security objectives and requirements of SRS Section 6.3. They run on the test environment with HTTPS enabled and use only dummy data.

Security validation activity	SEC ID	Method / tool	Pass criterion	Test case
Authentication enforcement on all endpoints	SEC-01	Postman/Newman requests without and with bad tokens	All protected endpoints return 401; only login is open	TC-SEC-01
Role-based authorization; privilege escalation attempts (attendant calling admin functions)	SEC-02	Scripted API calls per role; manual UI bypass	Forbidden calls return 403, even when the UI hides the control	TC-SEC-02, TC-AUTH-05
Password storage and logging check	SEC-03	Database inspection; log search	bcrypt hash, cost ≥ 10; no password in logs	TC-SEC-03

Security validation activity	SEC ID	Method / tool	Pass criterion	Test case
Transport security: TLS version and HTTP redirect	SEC-04	testssl.sh / SSL Labs	Only TLS 1.2+; HTTP redirected to HTTPS	TC-SEC-04
Fuzzing and injection: NoSQL operators, XSS, over-long and wrong-type input	SEC-05	OWASP ZAP baseline/active scan; manual payloads	Rejected with 400; no script runs; no data stored	TC-SEC-05
Session timeout and logout invalidation	SEC-06	Idle test; token reuse	401 after 15 minutes idle and after logout	TC-SEC-06, TC-AUTH-03
Secret scanning and dependency audit	SEC-07	gitleaks; npm audit	No secrets in repository; no known Critical vulnerabilities in dependencies	TC-SEC-07
Error-message and information-leak review	SEC-02, SEC-05	Manual review of error responses	No stack traces or database details in any error	TC-AUTH-02, TC-AUTH-05

6. Test Environment

Hardware: Developer laptops (8 GB RAM or more) for unit, integration and system tests; one server or cloud VM for the deployed test environment and performance runs; a smartphone or browser device mode for 360 px width.

Software: VPS v1.1 build (React.js frontend, Node.js LTS/Express.js backend), MongoDB (separate test database), Chrome, Firefox and Edge (latest two stable versions each).

Tools: Jest and Supertest (unit/integration), Cypress (UI automation), Postman/Newman (API), Apache JMeter (performance), OWASP ZAP and testssl.sh (security), gitleaks and npm audit (static security), GitHub Issues (defect tracking), Git (version control).

Test data: Dummy users of each role, vehicles of each type, slots of each type and status, and the default hourly rates (Appendix A). The performance database holds 500 vehicles and 100 slots. No real personal data is used.

7. Test Schedule

Proposed milestones; to be aligned with the course schedule.

Milestone	Date
Test plan and test case design	04-Oct-2026 to 07-Oct-2026
Environment setup and test data preparation	12-Oct-2026
Unit and integration test execution (with development)	13-Oct-2026 to 20-Oct-2026
System test execution (functional, usability, compatibility)	21-Oct-2026 to 30-Oct-2026
Security, performance and reliability testing	27-Oct-2026 to 31-Oct-2026
Regression testing and defect closure	02-Nov-2026 to 04-Nov-2026
UAT	05-Nov-2026 to 06-Nov-2026
Test summary report	07-Nov-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated) – Appendix B
- Test Scripts (Jest, Cypress, Postman collections, JMeter plans)
- Test Data – Appendix A
- Test Execution Logs
- Defect Reports
- Updated RTM (SRS Appendix C and Section 13)
- Test Summary Report

9. Roles and Responsibilities

Roles are proposed; every team member takes part in all activities.

Role	Name	Responsibility
QA Lead	Ramitha	Prepare the plan, coordinate execution, own authentication and security tests (TC-AUTH, TC-SEC)
Test Engineer	Snehitha	Design and execute test cases and log defects; own vehicle, slot, entry and exit tests (TC-VEH, TC-SLOT, TC-ENT, TC-EXT)
Test Engineer	Pramiti	Design and execute test cases and log defects; own fee, search, report, performance and usability tests (TC-FEE, TC-SRCH, TC-REP, TC-PERF, TC-USAB)
Developer	All team members	Support defect fixes and triage for their own modules
Product Owner	Course instructor / reviewer	Approve test results and sign off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds; test finished modules first
Test environment downtime	Keep a second environment on a local machine; keep database seed scripts
Hourly rates still unconfirmed	Test with the default rates (Appendix A); make expected values data-driven so only the data changes
Race conditions are hard to reproduce	Use a scripted parallel-request test repeated 50 times (TC-CONC-01)
Small team and limited time for security testing	Prioritise High-severity security tests; use automated scanners (ZAP, testssl.sh, gitleaks)
Performance results differ between laptop and server	Run performance tests on the deployed test environment and record its configuration

11. Assumptions & Dependencies

- The SRS requirements are stable; changes are reflected in the RTM before test execution.

- Default hourly rates are confirmed or replaced before system testing.
- Test data and test accounts are available before execution.
- The MongoDB test database and a TLS certificate for the test server are available.
- The Software Architecture & Design Specification is available for the role matrix used in TC-SEC-02.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours.
- The build is too unstable and blocks more than 30% of test cases.
- A Critical defect in login or entry/exit blocks all further testing.

Resume testing if:

- Blocking defects are resolved and the smoke test passes.
- The environment is stabilized.

13. Test Case Management & Traceability

The RTM ensures that every SRS requirement maps to one or more test cases. Test case IDs follow the pattern TC-<MODULE>-<NN> (AUTH, VEH, SLOT, ENT, EXT, FEE, SRCH, REP, PERF, CONC, REL, USAB, PORT, SEC). Detailed steps and expected results are in Appendix B. The "Test Case ID" columns of SRS Appendix C are to be updated with the IDs below.

Coverage summary: 37 test cases cover 42 requirement items (23 FR, 7 NFR, 7 SEC, 5 BR); requirement coverage is 100%.

Req. ID	Requirement	Priority	Test case IDs	Level of first test
FR-01	Authenticate user and issue session token	High	TC-AUTH-01	System
FR-02	Generic invalid-credentials message	High	TC-AUTH-02	System
FR-03	Logout ends the session	High	TC-AUTH-03	Integration
FR-04	Administrator manages user accounts and roles	High	TC-AUTH-04	Integration
FR-05	Deny out-of-role access (HTTP 403)	High	TC-AUTH-05	Integration
FR-06	Register a vehicle (format, upper case)	High	TC-VEH-01, TC-VEH-03	System
FR-07	Reject incomplete, invalid or duplicate vehicle	High	TC-VEH-02, TC-VEH-03	System
FR-08	Add a parking slot	High	TC-SLOT-01	System
FR-09	Set slot Unavailable / Available (not if Occupied)	Medium	TC-SLOT-02	Integration
FR-10	Display slot status and counts	High	TC-SLOT-03	System
FR-11	Record vehicle entry; reject if already parked	High	TC-ENT-01, TC-ENT-02	System
FR-12	Automatic or manual slot allocation by slot type	High	TC-ENT-01, TC-ENT-04	System
FR-13	Refuse entry when no matching slot	High	TC-ENT-03	System

Req. ID	Requirement	Priority	Test case IDs	Level of first test
FR-14	Atomic entry record with ticket; slot Occupied	High	TC-ENT-01	System
FR-15	Record exit and exit time; reject without active record	High	TC-EXT-01, TC-EXT-02	System
FR-16	Fee calculation with fixed 15-min grace and rounding	High	TC-EXT-03, TC-FEE-02	Unit
FR-17	Show fee summary before exit confirmation	High	TC-EXT-01	System
FR-18	Complete record and release slot	High	TC-EXT-01	System
FR-19	Configure hourly rate per vehicle type	Medium	TC-FEE-01, TC-FEE-02	System
FR-20	Search vehicle by registration number	High	TC-SRCH-01	System
FR-21	Current status Parked / Not Parked	High	TC-SRCH-02	System
FR-22	Parking history of a vehicle	Medium	TC-SRCH-03	System
FR-23	Parking Occupancy Report	Medium	TC-REP-01	System
NFR-01	Search response ≤ 3 s (95%, 50 users)	High	TC-PERF-01	Performance
NFR-02	Entry with allocation ≤ 3 s (95%)	High	TC-PERF-02	Performance
NFR-03	Exit with fee ≤ 3 s (95%)	High	TC-PERF-02	Performance
NFR-04	No double allocation under concurrency	High	TC-CONC-01	Integration
NFR-05	Acknowledged operations are persisted	High	TC-REL-01	System
NFR-06	Entry in ≤ 5 actions and ≤ 60 s	Medium	TC-USAB-01	Usability
NFR-07	Latest two Chrome/Firefox/Edge; ≥ 360 px	Medium	TC-PORT-01	System
SEC-01	Authentication before protected functions	High	TC-AUTH-01, TC-SEC-01	System
SEC-02	Server-side role-based authorization	High	TC-AUTH-05, TC-SEC-02	Integration
SEC-03	Passwords as salted bcrypt (cost ≥ 10)	High	TC-SEC-03	Security
SEC-04	HTTPS with TLS 1.2 or higher	High	TC-SEC-04	Security
SEC-05	Server-side input validation	High	TC-SEC-05	Security
SEC-06	Session timeout after 15 min idle	Medium	TC-SEC-06	Security
SEC-07	Secrets outside source code	Medium	TC-SEC-07	Security
BR-01	One active record per slot	High	TC-CONC-01	Integration
BR-02	Allocate only Available, type-matching slots	High	TC-ENT-01	System
BR-03	One active record per vehicle; exit needs active record	High	TC-ENT-02, TC-EXT-02	Integration
BR-04	Fees use admin-set rates and fixed grace period	High	TC-EXT-03	Unit
BR-05	Only Administrator manages slots, users, rates, reports	High	TC-AUTH-05, TC-REP-01	Integration

Examples: FR-16 (fee calculation) → TC-EXT-03, TC-FEE-02; SEC-06 (session timeout) → TC-SEC-06; NFR-01 (search response time) → TC-PERF-01.

14. Test Metrics & Reporting

Metrics collected:

- % of test cases executed
- % passed / failed / blocked
- Defect density (defects per module)
- Defect aging (days open by severity)
- Requirement coverage (% of requirements with at least one passed test)
- Code coverage of backend service functions (reported for information)
- Performance results: p95 response time per scenario

Reports:

- Daily execution status (during system test)
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Ramitha	
Dev Lead		
Product Owner		

Appendix A: Test Data

Item	Data
Users	admin1 (Administrator); att1, att2 (Parking Attendant); deactivated user "old1"
Vehicles	KA01AB1234 (Car); KA02CD5678 (Car); KA03EF9012 (Two-wheeler); KA04GH3456 (Other); unregistered number KA09ZZ0000
Slots	Car: A-01 to A-10; Two-wheeler: B-01 to B-05; Other: C-01 to C-02
Hourly rates (default)	Two-wheeler ₹10; Car ₹30; Other ₹50; fixed grace period 15 minutes; billable hours = (duration – 15 min) rounded up
Fee examples (Car)	10 min ₹0; 15 min ₹0; 16 min ₹30; 75 min ₹30; 76 min ₹60; 195 min ₹90
Invalid inputs	Registration numbers "ABC", 16 characters, "KA@01#AB"; rate -5 and "abc"; payloads {"\$ne":""} and <script>alert(1)</script>
Performance data	500 vehicles, 100 slots, 50 virtual users, 10-minute runs

Appendix B: Test Cases

This appendix lists 37 test cases covering the functional, non-functional and security requirements. Type shows the test level and kind; Priority is High, Medium or Low. All expected results can be checked without interpretation.

TC ID	Title / Type / Priority	Req. IDs	Preconditions	Test steps	Test data	Expected result
TC-AUTH-01	Login with valid credentials System / Functional High	FR-01, SEC-01	Attendant account "att1" exists and is active.	1. Open the Login page. 2. Enter username and password of att1. 3. Click Login.	att1 / correct password	- A session token is issued and the attendant home page opens. - The page shows the username and role "Parking Attendant".
TC-AUTH-02	Login with invalid credentials System / Negative High	FR-02	Account "att1" exists.	1. Login with correct username and wrong password. 2. Login with an unknown username and any password.	att1 / wrong; nouser / x	- Both attempts show exactly "Invalid username or password". - No token is issued and the message does not say which field was wrong.
TC-AUTH-03	Logout invalidates the session Integration / Functional High	FR-03	User is logged in; token T is saved.	1. Click Logout. 2. Send a protected API request with token T.	Token T	- Logout returns the user to the Login page. - The API request with T is rejected with HTTP 401.
TC-AUTH-04	Administrator manages user accounts Integration / Functional High	FR-04	Administrator is logged in.	1. Create an Attendant account. 2. Change its role to Administrator. 3. Deactivate the account. 4. Try to login with the deactivated account.	att2 / Pass@1234	- Account is created, role is updated and account is deactivated as requested. - Login with the deactivated account fails.
TC-AUTH-05	Role-based denial of admin functions Integration / Security High	FR-05, SEC-02, BR-05	An Attendant token is available.	1. Using the attendant token call POST /slots (add slot). 2. Using the attendant token call PUT /fee-rates. 3. Using the attendant token call POST /users. 4. Using the attendant token call GET /reports/occupancy.	Attendant token	- Each request returns HTTP 403. - The response body contains no slot, fee-rate, user or report data.
TC-VEH-01	Register a vehicle System / Functional High	FR-06	Logged in as Attendant.	1. Open Vehicle Registration. 2. Enter registration number "ka01ab1234" and type Car. 3. Submit.	ka01ab1234, Car	- Vehicle is created, linked to the owner and confirmed on screen. - The registration number is stored as KA01AB1234.
TC-VEH-02	Reject incomplete or duplicate vehicle System / Negative High	FR-07	KA01AB1234 is already registered.	1. Submit with the vehicle type empty. 2. Submit with registration number KA01AB1234 again.	type empty; KA01AB1234	- Both submissions are rejected and no record is created. - The error message names the empty field and the duplicate registration number.

TC ID	Title / Type / Priority	Req. IDs	Preconditions	Test steps	Test data	Expected result
TC-VEH-03	Registration number format boundaries Unit / Boundary Medium	FR-06, FR-07	Validation function or form available.	1. Submit numbers of 3, 4, 15 and 16 characters. 2. Submit "KA 01-AB 1234" and "KA@01#AB".	"ABC", "ABCD", 15 chars, 16 chars, "KA 01-AB 1234", "KA@01#AB"	• 4 and 15 characters are accepted; 3 and 16 are rejected. • Hyphens and spaces are accepted; "@" and "#" are rejected.
TC-SLOT-01	Administrator adds a parking slot System / Functional High	FR-08	Administrator is logged in.	1. Add slot A-01 with type Car. 2. Add slot A-01 again.	A-01, Car	• First slot is created with status Available. • The duplicate Slot ID is rejected.
TC-SLOT-02	Set a slot Unavailable and back to Available Integration / Functional + Negative Medium	FR-09	Slot A-02 is Available; slot A-03 is Occupied.	1. Set A-02 to Unavailable. 2. Try to allocate A-02 to an entering car. 3. Set A-02 back to Available. 4. Set A-03 to Unavailable.	A-02, A-03	• A-02 becomes Unavailable and is not allocated; after being set back it is Available again. • Setting the Occupied slot A-03 to Unavailable is rejected.
TC-SLOT-03	Slot availability display and counts System / Functional High	FR-10	Facility has 10 Car slots (6 Available, 3 Occupied, 1 Unavailable) and 5 Two-wheeler slots (all Available).	1. Login as Attendant and open Slot Availability. 2. Compare displayed slots and counts with the database.	Seeded data	• Every slot is shown with Slot ID, type and correct status. • Counts shown: Car 6 Available / 3 Occupied; Two-wheeler 5 Available / 0 Occupied.
TC-ENT-01	Record a vehicle entry (happy path) System / Functional High	FR-11, FR-12, FR-14, BR-02	Car KA01AB1234 registered, not parked; Car slots A-01 and A-04 Available.	1. Login as Attendant. 2. Select the vehicle and click Record Entry.	KA01AB1234	• Slot A-01 (lowest Slot ID of matching type) is allocated. • A parking record with entry time, slot and a unique ticket number is created and the slot becomes Occupied.
TC-ENT-02	Reject entry for an already parked vehicle Integration / Negative High	FR-11, BR-03	KA01AB1234 has an Active parking record.	1. Record entry for KA01AB1234 again.	KA01AB1234	• Entry is rejected with a message that the vehicle is already parked. • No second record is created and no slot changes state.
TC-ENT-03	No matching slot available System / Negative High	FR-13	All Car slots are Occupied or Unavailable; Two-wheeler slots are free.	1. Record entry for a Car.	A car without a parking record	• Entry is refused with "No parking slot available for this vehicle type". • No parking record is created.
TC-ENT-04	Manual slot selection by the attendant System / Functional + Negative Medium	FR-12	Car slots A-04, A-05 Available; Two-wheeler slot B-01 Available.	1. Record entry for a Car and choose slot A-05. 2. Record entry for another Car and try to choose B-01.	Two cars; A-05; B-01	• First car is assigned A-05. • Choosing the Two-wheeler slot B-01 for a Car is rejected.
TC-EXT-01	Record a vehicle exit (happy path) System / Functional High	FR-15, FR-17, FR-18	Car KA01AB1234 has an Active record in slot A-01, entered 90 minutes ago.	1. Login as Attendant. 2. Search the vehicle and click Record Exit. 3. Check the summary screen. 4. Click Confirm Exit.	KA01AB1234	• Exit time (server time) is stored; the summary shows entry time, exit time, duration and fee. • After confirmation the record is Completed with the fee and slot A-01 becomes Available.

TC ID	Title / Type / Priority	Req. IDs	Preconditions	Test steps	Test data	Expected result
TC-EXT-02	Reject exit without an active record Integration / Negative High	FR-15, BR-03	Vehicle KA02CD5678 is registered and not parked.	1. Try to record exit for KA02CD5678.	KA02CD5678	- Exit is rejected with a message that no active parking record exists. - No data changes.
TC-EXT-03	Fee calculation boundary values Unit / Boundary High	FR-16, BR-04	Hourly rates: Two-wheeler ₹10, Car ₹30, Other ₹50; fixed grace period 15 minutes.	1. For a Car compute the fee for each duration. 2. For a Two-wheeler compute the fee for 76 minutes.	Car: 10, 15, 16, 75, 76, 195 min; Two-wheeler: 76 min	- Car fees: 10 min ₹0; 15 min ₹0; 16 min ₹30; 75 min ₹30; 76 min ₹60; 195 min ₹90. - Two-wheeler 76 min = ₹20.
TC-FEE-01	Configure hourly rates System / Functional + Negative High	FR-19	Administrator is logged in.	1. Set Car rate to 40. 2. Set Car rate to -5. 3. Set Car rate to "abc".	40; -5; abc	40 is saved and displayed. -5 and "abc" are rejected; the stored rate remains 40.
TC-FEE-02	Rate change applies only to later exits Integration / Functional Medium	FR-19, FR-16	Car C1 is parked (Active record) with Car rate ₹30.	1. Change the Car rate to ₹40. 2. Record exit of C1 after 2 hours. 3. Record entry and a 2-hour exit of car C2.	Rate 30 → 40	- Both C1 and C2 are billed at ₹40 because their exit is recorded after the change. - A Completed record keeps its stored fee when the rate is changed again.
TC-SRCH-01	Search a vehicle by registration number System / Functional + Negative High	FR-20	Vehicles KA01AB1234 and KA02CD5678 are registered.	1. As Attendant search KA01AB1234. 2. As Administrator search KA02CD5678. 3. Search the unregistered number KA09ZZ0000.	KA01AB1234, KA02CD5678, KA09ZZ0000	- Both registered vehicles are found with their details. - The unregistered number returns "Vehicle not found" and no data.
TC-SRCH-02	Current status of a vehicle System / Functional High	FR-21	One vehicle is parked in A-01; one is not parked.	1. Search the parked vehicle. 2. Search the non-parked vehicle.	Parked and non-parked vehicles	- Parked vehicle: status Parked with Slot ID A-01, entry time and elapsed duration. - Non-parked vehicle: status Not Parked.
TC-SRCH-03	Parking history of a vehicle System / Functional Medium	FR-22	A vehicle has 3 Completed records.	1. Open the history of the vehicle.	3 records	- Three records are shown with entry time, exit time, Slot ID and fee. - Order is most recent first.
TC-REP-01	Parking occupancy report System / Functional Medium	FR-23, BR-05	Seeded slots with a mix of statuses.	1. As Administrator generate the Parking Occupancy Report. 2. Compare it with the database. 3. As Attendant request the same report.	Seeded data	- Each slot is listed with status, assigned vehicle and last updated time, and values match the database. - The attendant request is denied (HTTP 403).
TC-PERF-01	Vehicle search response time Performance High	NFR-01	Database seeded with 500 vehicles and 100 slots; JMeter script ready.	1. Run 50 concurrent virtual users for 10 minutes, each searching random registration numbers. 2. Calculate the 95th percentile response time.	50 users; 10 min; ramp-up 30 s	95th percentile response time is 3 seconds or less. - Error rate is below 1%.

TC ID	Title / Type / Priority	Req. IDs	Preconditions	Test steps	Test data	Expected result
TC-PERF-02	Entry and exit response time Performance High	NFR-02, NFR-03	Enough free slots and parked vehicles for 200 operations.	1. Run 20 concurrent users, each performing entry then exit, 200 operations in total. 2. Calculate the 95th percentile for entry and for exit separately.	20 users; 200 operations	• 95th percentile of entry (with allocation) is 3 seconds or less. • 95th percentile of exit (with fee calculation) is 3 seconds or less.
TC-CONC-01	Concurrent allocation of the last slot Integration / Concurrency High	NFR-04, BR-01	Exactly one Car slot is Available; two different cars are registered.	1. Send two entry requests for the two cars at the same instant from a script. 2. Repeat the run 50 times, resetting the data each time.	2 cars; 1 slot; 50 runs	• In every run exactly one request succeeds and the other is told to retry or that no slot is available. • The slot never has two Active records.
TC-REL-01	Data persistence after restart System / Reliability High	NFR-05	Entry and exit operations have just been acknowledged as successful.	1. Record 5 entries and 3 exits and note the success responses. 2. Restart the backend service and the MongoDB service. 3. Query all records.	5 entries; 3 exits	• All 8 acknowledged operations are present with the same values.
TC-USAB-01	Usability of vehicle entry Usability Medium	NFR-06	Three first-time users briefly trained (10 minutes).	1. Each of the three users records one vehicle entry; count the user actions and the time taken.	3 participants	• Every entry takes 5 user actions or fewer and under 60 seconds.
TC-PORT-01	Browser and screen-size compatibility System / Compatibility Medium	NFR-07	Latest two stable versions of Chrome, Firefox and Edge installed.	1. Run the smoke suite (login, slot view, entry, exit) in each browser. 2. Repeat at screen widths 360, 768 and 1366 px.	3 browsers × 2 versions; 3 widths	• All smoke steps pass in every browser. • No horizontal scrolling or overlapping controls at 360 px.
TC-SEC-01	Authentication required Security / API High	SEC-01	API running.	1. Call every protected endpoint with no token and with a malformed token. 2. Open the Slot Availability page without logging in.	No token; "abc"	• All protected endpoints return HTTP 401. • Only the login function is accessible without login.
TC-SEC-02	Role-based authorization matrix Security / API High	SEC-02	One user of each role.	1. For each role call every endpoint listed in the role matrix of the SAD. 2. Compare actual with expected HTTP status.	3 roles × all endpoints	• Permitted calls succeed; all others return 403. • Authorization is enforced by the server even when the UI hides the control.
TC-SEC-03	Password storage Security / Inspection High	SEC-03	Test user created.	1. Read the stored password field in MongoDB. 2. Search application logs for the test password.	Password "Test@1234"	• Field is a bcrypt hash (starts with \$2a\$/\$2b\$) with cost 10 or higher and a unique salt. • The password does not appear in any log.
TC-SEC-04	Transport security (HTTPS / TLS) Security / Scan High	SEC-04	Deployed test environment with a certificate.	1. Scan the server with testssl.sh or SSL Labs. 2. Request the site over plain HTTP.	Test server URL	• Only TLS 1.2 and higher are accepted. • HTTP is redirected to HTTPS.

TC ID	Title / Type / Priority	Req. IDs	Preconditions	Test steps	Test data	Expected result
TC-SEC-05	Input validation: injection and XSS Security / Fuzzing High	SEC-05	API and UI running.	1. Send {"\$ne":""} and {"\$gt":""} as username and password. 2. Enter <script>alert(1)</script> in registration number and name fields. 3. Send over-long (10,000 characters) and wrong-type values.	NoSQL operators; script payload; long strings	• All malicious input is rejected with HTTP 400. • No script is executed and no data is stored.
TC-SEC-06	Session timeout Security / Functional Medium	SEC-06	User is logged in.	1. Stay idle for 15 minutes and 30 seconds. 2. Perform any action.	Idle 15 min 30 s	• The request is rejected with HTTP 401 and the user is returned to the Login page.
TC-SEC-07	Secrets are not in source code Security / Static Medium	SEC-07	Repository available.	1. Run gitleaks (or git-secrets) on the full history. 2. Check that DB URI and token secret are read from environment variables.	Repository	• No credentials or signing secrets are found in the repository. • The application fails to start without the environment configuration.